Retaliatory Evictions: Distributed-Lag Evidence

Philly Evictions Project

2026-03-27

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1 Overview

This writeup now reports only the retaliatory timing analysis from `r/retaliatory-evictions.r`:

- distributed-lag event-time regressions (`filed_eviction` on lags/leads of complaint indicators)
- bandwidth diagnostics around eviction filings
- distributed-lag heterogeneity by building-level `% Black` split

Complaint-suppression analysis was intentionally removed from this workflow and now lives in `r/comp-statics-annual-race.R` and `writeups/comp-statics-annual-race.qmd`. Unless explicitly overridden in the script call, outputs shown here are pre-COVID only (`<= 2019`).

2 Empirical Bayes Filing-Rate Memo

The maintenance regressions below now use `filing_rate_eb_pre_covid` as the default filing-intensity measure. This is the building-level Empirical Bayes (Gamma-Poisson) posterior mean pooled over pre-2019 years:

$$y_{it} \mid \lambda_i \sim \text{Poisson}(E_{it}\lambda_i), \qquad \lambda_i \sim \text{Gamma}(\alpha, \beta),$$

with sufficient statistics

$$Y_i = \sum_t y_{it}, \qquad E_i = \sum_t E_{it},$$

and posterior mean

$$\mathbb{E}[\lambda_i \mid Y_i, E_i] = \frac{\alpha + Y_i}{\beta + E_i}.$$

In the production pipeline (`r/make-analytic-sample.R`, section 6b), the city prior is estimated by method-of-moments; ZIP-specific priors are used when supported, otherwise the city prior is used as fallback. The raw long-run filing rate remains available for robustness and is reported in the appendix of this writeup.

Reference memo for reuse: `docs/notes/eb-filing-rate-memo-2026-03-04.md`.

3 Estimating Equations

3.1 Distributed lag (main)

For complaint stem $c \in \{\text{complaint}, \text{severe}, \text{non_severe}\}$:

$$Y_{it} = \sum_{k=1}^{h} \beta_{k,c}^{-} D_{i,t-k}^{c} + \beta_{c}^{0} D_{i,t}^{c} + \sum_{k=1}^{h} \beta_{k,c}^{+} D_{i,t+k}^{c} + \alpha_i + \gamma_t + \varepsilon_{it},$$

where:

- $Y_{it} = \mathbb{1}[\text{filed_eviction}_{it} = 1]$
- $D_{it}^{c} = \mathbb{1}[\text{filed_}c_{it} = 1]$
- α_i: PID fixed effects, γ_t: period fixed effects
- SEs clustered at PID

3.2 Race heterogeneity split

Define:

$$G_{it} = \mathbb{1}[\text{infousa_pct_black}_{it} \geq c_{\text{pre}}],$$

where c_{pre} is computed from pre-2020 $(PID, year)$-unique observations using the configured rule (default: pre-COVID mean).

The same distributed-lag equation is estimated on high- and low-G_{it} subsamples.

4 Data and Sample

- Unit: PID × quarter
- Default window: 2011–2019
- Inputs: rental building panel, analytic PID filter, eviction filings + address crosswalk, and `bldg_panel_blp` tenant composition
- Eviction filing sample restrictions (implemented in `r/retaliatory-evictions.r`):
  - keep only filings with unique parcel crosswalk matches (`num_parcels_matched == 1`)
  - drop commercial filings (`commercial != TRUE`)
  - drop non-residential filings (`non_residential != TRUE`)
  - keep only non-payment-of-rent filings (`a == TRUE` in `evictions_clean`)
  - apply the configured year window (default pre-COVID: 2011–2019)
- `Full` complaint indicator: any complaint filing in the period.
- `Severe` complaint indicator: any period complaint in the severe bundle (`heat`, `fire`, or `property_maintenance`), consistent with `total_severe_complaints` in the panel.
- `Non-Severe` complaint indicator: any complaint in the non-severe bundle used in the script.

This aligns the filing sample with the non-payment focus in Lodermeier (2025) and with the eviction data README note that complaint-derived fields are consistently available from 2006 onward, with a complaint format change in 2011.

Table 1: Eviction filing sample restrictions used in this analysis

Step	Rows	Share of raw
raw_evictions_clean_rows	796,440	100.0%
unique_parcel_xwalk_match	533,882	67.0%
drop_commercial_cases	532,945	66.9%
drop_non_residential_cases	528,750	66.4%
keep_nonpayment_of_rent_cases_a_flag	338,850	42.5%
keep_year_between_2011_and_2019	176,174	22.1%

5 Bandwidth Diagnostics

Table 2: Eviction filing decomposition within quarters windows

Bandwidth	Same period	Within bandwidth	Within bandwidth, not same period	Non-retaliatory
1	14.0%	26.3%	12.3%	73.7%
2	14.0%	32.8%	18.8%	67.2%

6 Distributed-Lag Results

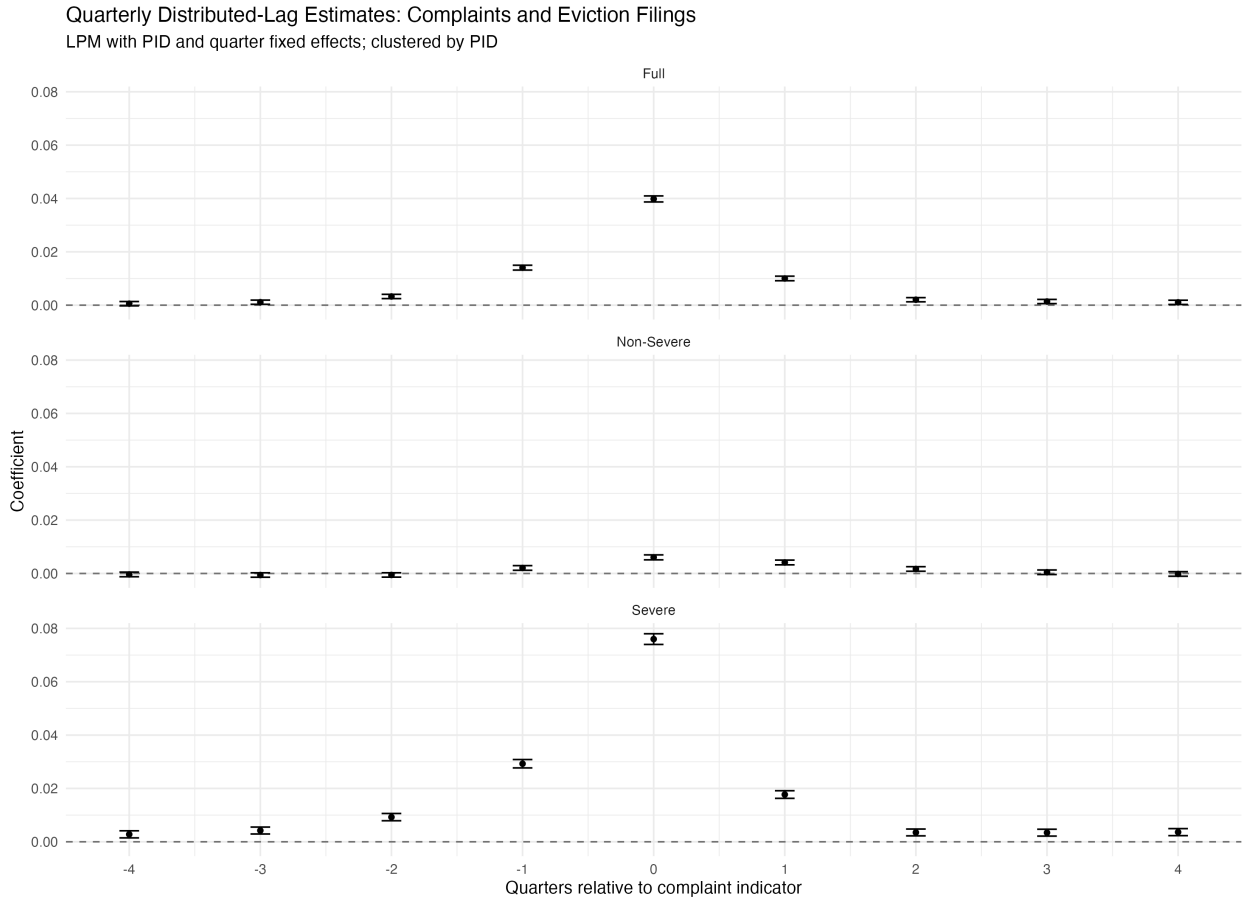

Figure 1: Distributed-lag coefficients (pre-COVID): effect of complaint indicators at lag/lead k on eviction filing indicator. Panels are labeled Full, Non-Severe, and Severe. PID and period FE; SE clustered by PID.

7 % Black Heterogeneity in Distributed Lags

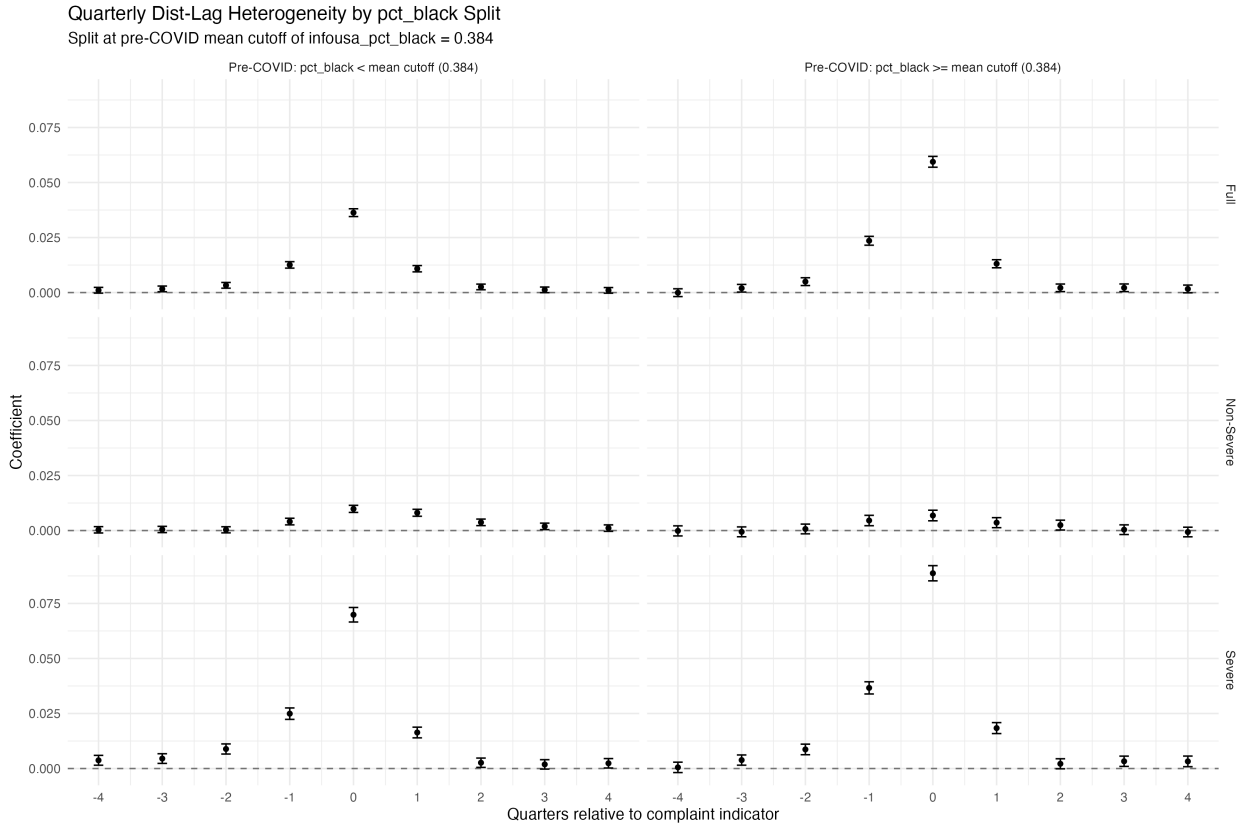

Figure 2: Distributed-lag coefficients by high/low pct black split (cutoff from pre-COVID distribution rule configured in script).

8 Portfolio Eviction Group Heterogeneity

Buildings are assigned to one of three portfolio eviction groups based on a **leave-one-out (LOO)** EB pre-COVID filing rate from the other buildings in the same conglomerate:

- **Solo**: no other buildings in the conglomerate (LOO rate is undefined)
- **Low-evicting portfolio**: LOO rate below the median across multi-building conglomerates
- **High-evicting portfolio**: LOO rate at or above the median

This framing asks whether retaliation is more common, more severe, and maintenance less responsive *when a building's landlord operates other high-evicting properties* — isolating a portfolio-level behavioral pattern rather than just the focal building's own filing history.

8.1 Distributed-Lag Results by Portfolio Eviction Group

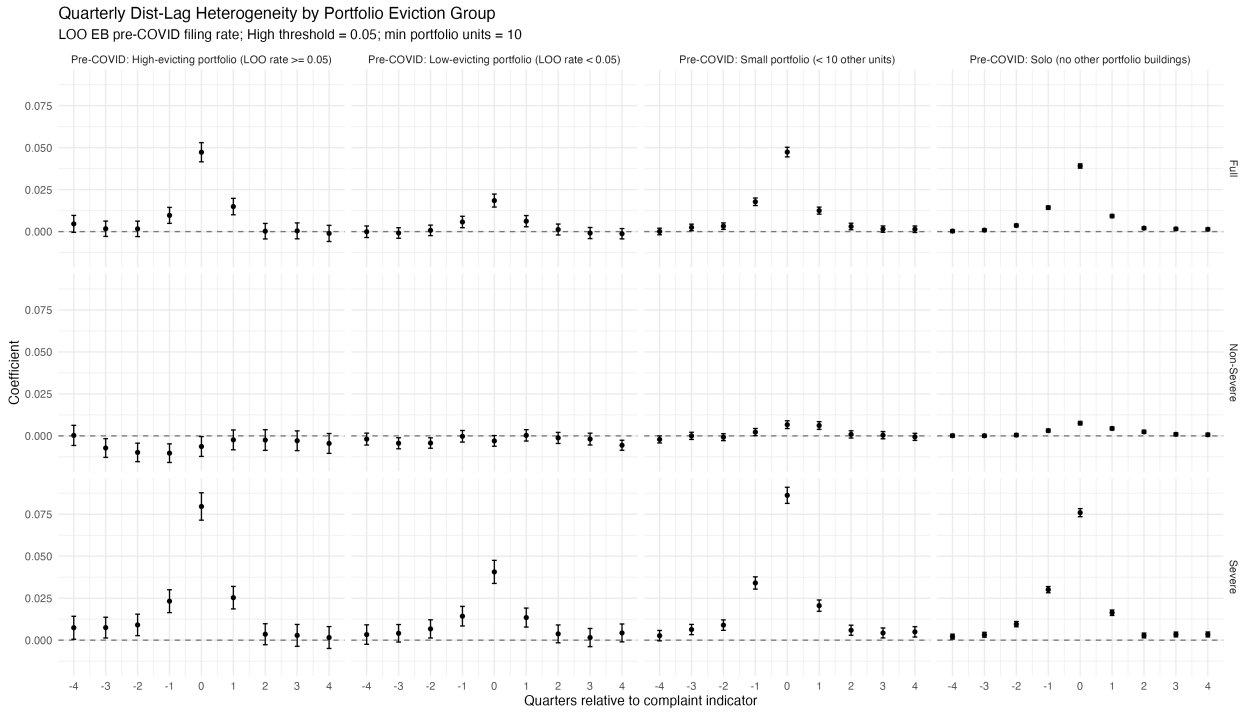

Figure 3: Distributed-lag coefficients by portfolio eviction group. Three groups: Solo (no conglomerate partners), Low-evicting portfolio (LOO EB rate below median), High-evicting portfolio (LOO EB rate at/above median). PID and period FE; SE clustered by PID.

9 One-Month Back-Rent Regressions

Case assignment for this block:

- Outcome: `one_month_back_rent = 1` when `total_rent / ongoing_rent` is within `1 ± tol` (default `tol = 0.10`).
- `retaliatory_severe = 1` if an eviction case's PID-period has a severe complaint in the same period or within `± bandwidth` periods (default bandwidth is `2` quarters).
- `high_black_split = 1` if building `infousa_pct_black` is above the pre-COVID mean cutoff.
- Defendant race category from case-level imputation (`r/impute-race.R` output): `black` if `case_p_black >= 0.8`, `white` if `case_p_white >= 0.8`, and `unknown` otherwise.
- In the defendant-race interaction model, `white` is the reference category.

Table 3: Pre-COVID case assignment by severe-complaint proximity

Status	Cases	Share
non_retaliatory	86927	61.6%
same_period_severe	25675	18.2%
within_bw_severe_not_same_period	28518	20.2%

Table 4: Regression sample summary (pre-COVID; PID + year FE specs)

Spec	N	Mean(1 month rent)	Same-period severe share	Within-bw severe (not same period) share	Above-mean pct black share	Defendant race: Black	Defendant race: White	Defendant race: Unknown	Defendant race impute-ok share
Status dummies only	115491	0.1890	18.2%	20.2%	55.5%	–	–	–	99.5%
Status x building above-mean pct black	93349	0.1872	18.4%	21.2%	55.5%	–	–	–	99.6%

Dependent Variable:	one_month_back_rent	
Model:	(1)	(2)
Variables		
Same-period severe	0.017**	-0.007
	(0.006)	(0.013)
One-quarter pre/post severe	0.0003	-0.013
	(0.006)	(0.010)
Above-mean building pct black		-0.003
		(0.009)
Same-period severe x above-mean building pct black		0.039**
		(0.017)
One-quarter pre/post severe x above-mean building pct black		0.026**
		(0.013)
% cases filed in quarter of complaint	18.2	18.4
% cases in one-quarter pre/post window	20.2	21.2
Fixed-effects		
PID	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	115,491	93,349
R^2	0.28980	0.29007
Within R^2	0.00023	0.00048

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9.1 Poisson (PPML) Regressions: Months Back Rent

The same three specifications are repeated using Poisson (PPML) with `months_back_rent` (= `total_rent / ongoing_rent`, continuous) as the outcome. Coefficients are log-multiplicative; SEs clustered by PID.

Dependent Variable:	months_back_rent	
Model:	(1)	(2)
Variables		
Same-period severe	-0.027**	0.017
	(0.013)	(0.025)
One-quarter pre/post severe	0.004	0.025
	(0.010)	(0.017)
Above-mean building pct black		0.023
		(0.017)
Same-period severe x above-mean building pct black		-0.064**
		(0.030)
One-quarter pre/post severe x above-mean building pct black		-0.036*
		(0.021)
% cases filed in quarter of complaint	18.2	18.4
% cases in one-quarter pre/post window	20.2	21.2
Fixed-effects		
PID	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	115,427	93,293
Squared Correlation	0.48618	0.49386
Pseudo R^2	0.16446	0.16349
BIC	607,408.0	482,032.2

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9.2 OLS Log-Rent Regression: log(Total Rent) | log(Ongoing Rent)

Outcome is `log(total_rent)` with `log(ongoing_rent)` entered as a continuous covariate. Conditioning on log ongoing rent removes the mechanical correlation between total back rent and rent level across tenants. The coefficient on `retaliatory_status` therefore captures the percentage difference in total back rent *claimed* for retaliatory versus non-retaliatory cases, holding the monthly rent fixed. A negative coefficient means landlords file for less back rent in retaliatory cases — i.e., the filing is not primarily about recovering arrears.

Dependent Variable:	log(total_rent)	
Model:	(1)	(2)
Variables		
Same-period severe	-0.024**	0.014
	(0.012)	(0.023)
One-quarter pre/post severe	0.004	0.028*
	(0.008)	(0.015)
log(ongoing_rent)	0.692***	0.706***
	(0.019)	(0.021)
Above-mean building pct black		0.008
		(0.015)
Same-period severe x above-mean building pct black		-0.056**
		(0.027)
One-quarter pre/post severe x above-mean building pct black		-0.045**
		(0.019)
% cases filed in quarter of complaint	18.2	18.4
% cases in one-quarter pre/post window	20.2	21.2
Fixed-effects		
PID	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	113,104	91,418
R^2	0.49717	0.50158
Within R^2	0.04082	0.04247

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9.3 Back-Rent Regressions by Portfolio Eviction Group

Separate LPM (`one_month_back_rent`) and PPML (`months_back_rent`) regressions run for each portfolio eviction group. Specification mirrors the main models above (PID + year FE; SE clustered by PID; pre-COVID sample). This tests whether **high-evicting-portfolio landlords are more likely to file retaliatory cases with precisely one month of back rent claimed**.

Table 5: LPM: one-month back rent retaliatory status, by portfolio eviction group (pre-COVID; PID + year FE)

Group	Term	Est	SE	p-value	N
Solo	same_period_severe	0.0338	0.0079	0.000	58749
Solo	within_bw_severe_not_same_period	0.0042	0.0071	0.552	58749
Small portfolio	same_period_severe	0.0088	0.0123	0.475	16356
Small portfolio	within_bw_severe_not_same_period	0.0004	0.0118	0.970	16356
Low-evicting portfolio	same_period_severe	0.0261	0.0246	0.290	8553
Low-evicting portfolio	within_bw_severe_not_same_period	-0.0025	0.0194	0.898	8553
High-evicting portfolio	same_period_severe	-0.0079	0.0137	0.566	31833
High-evicting portfolio	within_bw_severe_not_same_period	-0.0050	0.0124	0.683	31833

10 Equilibrium Maintenance Analysis

This section examines whether landlords adjust maintenance investment (proxied by permit activity) in response to eviction pressure and tenant composition, and whether permit activity responds dynamically to severe complaints.

10.1 Baseline Maintenance Permit Regressions

Poisson (PPML) regressions of `total_permits` on building-level eviction filing intensity and racial composition, with `log(total_units)` as offset. Main specifications use EB pre-COVID filing intensity (`filing_rate_eb_pre_covid`) with:

- a high-filing dummy (`>= 15%`)
- filing bins: `[0,1)`, `[1,5)`, `[5,10)`, `[10,20)`, `[20,inf)` (all on EB rates)

The rent-augmented variants add mean inflation-adjusted log rent (within PID over 2011–2019) to partial out rent-driven maintenance incentives.

Dependent Variable:	total_permits			
Model:	(1)	(2)	(3)	(4)
Variables				
above_mean_pct_black_btTRUE	-0.409***	-0.347***	-0.404***	-0.339***
	(0.007)	(0.009)	(0.007)	(0.009)
high_filing_eb	-0.119***	-0.044***		
	(0.011)	(0.012)		
mean_log_med_rent_inf_adj_PID		0.419***		0.406***
		(0.010)		(0.010)
filing_bin_eb = [1,5)")			0.198***	0.046***
			(0.009)	(0.010)
filing_bin_eb = [5,10)")			-0.199***	-0.174***
			(0.009)	(0.010)
filing_bin_eb = [10,20)")			-0.148***	-0.098***
			(0.011)	(0.012)
filing_bin_eb = [20,inf)")			-0.174***	-0.130***
			(0.015)	(0.015)
Fixed-effects				
building_type	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
quality_grade	Yes	Yes	Yes	Yes
year_blt_decade	Yes	Yes	Yes	Yes
num_units_bin	Yes	Yes	Yes	Yes
GEOID	Yes	Yes	Yes	Yes
has_rent_data	Yes		Yes	
Fit statistics				
Observations	1,683,222	749,684	1,673,765	748,855
Squared Correlation	0.30725	0.23903	0.30450	0.24125
Pseudo R^2	0.20489	0.25429	0.20529	0.25476
BIC	1,133,920.2	609,995.0	1,126,208.3	607,192.0

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

10.2 Baseline Maintenance Regressions by Portfolio Eviction Group

PPML regressions of `total_permits` on racial composition and **LOO portfolio eviction group** (`portfolio_evict_group`; reference = "Solo"), with and without the EB high-filing dummy. LOO classification uses a 0.05 threshold on other conglomerate buildings' EB pre-COVID filing rate (10 other units = "High-evicting portfolio"; < 10 other units = "Small portfolio").

Dependent Variable:	total_permits	
Model:	(1)	(2)
Variables		
above_mean_pct_black_btTRUE	-0.415***	-0.411***
	(0.007)	(0.007)
portfolio_evict_groupSmallportfolio	-0.091***	-0.089***
	(0.007)	(0.007)
portfolio_evict_groupLow-evictingportfolio	0.047***	0.044***
	(0.008)	(0.008)
portfolio_evict_groupHigh-evictingportfolio	0.042***	0.063***
	(0.011)	(0.011)
high_filing_eb		-0.125***
		(0.011)
Fixed-effects		
building_type	Yes	Yes
year	Yes	Yes
quality_grade	Yes	Yes
year_blt_decade	Yes	Yes
num_units_bin	Yes	Yes
GEOID	Yes	Yes
has_rent_data	Yes	Yes
Fit statistics		
Observations	1,683,222	1,683,222
Squared Correlation	0.30549	0.30558
Pseudo R^2	0.20498	0.20507
BIC	1,133,823.1	1,133,715.3

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

10.3 Spell-Level Maintenance Regressions

Each observation is a **PID × ownership-spell** — a contiguous block of years during which the same conglomerate owned the parcel (2011–2019). This collapses the ~1.6M-row annual panel to ~100–250 K spells, making the cross-sectional unit honest and covariates spell-specific (filing rate, rent, and % Black are averages during the landlord's tenure).

Design notes:

- **Filing intensity** (`high_filing_spell`): mean of `filing_rate_eb` (pooled pre-COVID EB rate) over spell years. Because this EB rate is a static PID-level scalar, the spell mean equals the PID EB rate. Spell-varying filing intensity is available via the raw rate robustness column.
- **FE spec**: `building_type + start_year + GEOID`. `quality_grade`, `year_blt_decade`, and `num_units_bin` are included as RHS covariates rather than FEs. The 6-way FE spec from the annual model is too sparse at spell level (~267K spells vs ~72M possible cells → all PPML singletons); this 3-way spec gives ~14K cells and ~19 spells/cell. `start_year` absorbs permit trends by ownership-entry cohort.
- **M4 (PID FE)**: identifies from within-parcel ownership transitions. `high_filing_spell` is collinear with PID FE (EB rate does not vary by year) and is dropped automatically by `fixest`. Identification comes from `portfolio_evict_group_cong` and `above_mean_pct_black_spell`, which vary across spells if conglomerate composition or pct_Black changes at ownership transition.
- **LOO classification** (`portfolio_evict_group_cong`): computed from the **full 2011–2019 panel** (not the spell window) via `compute_loo_filing_type()`. Landlord type is a structural attribute; full-period averaging reduces noise. Reference = "Solo". Threshold: 0.05 filings/unit-year; small-portfolio screen: `loo_min_units × 9` unit-years.

Table 6: Spell-level maintenance sample summary

n_spells	n_pid	n_neg_spells	n_total_pid	n_total_multi_pid	mean_spell_years	median_spell_years	mean_unit_years	pct_high_filing	has_rent
267002	189554	267296	189848	57703	6.3	8	12.2	0.0725	0.3405

Dependent Variable:	total_permits_sum			
Model:	(1)	(2)	(3)	(4)
Variables				
above_mean_pct_black_spellTRUE	-0.083***	-0.030***	-0.115***	
	(0.006)	(0.009)	(0.0008)	
high_filing_spell	-0.127***	-0.068***	-0.137***	
	(0.011)	(0.013)	(0.002)	
has_rent_data_spellTRUE	0.200***		0.392***	
	(0.006)		(0.001)	
year_blt_decade1910	0.300***	0.229***	0.870***	
	(0.015)	(0.019)	(0.002)	
year_blt_decade1920	0.143***	0.091***	0.132***	
	(0.014)	(0.017)	(0.002)	
year_blt_decade1930	0.100***	0.024	0.271***	
	(0.019)	(0.024)	(0.003)	
year_blt_decade1940	0.155***	0.150***	0.437***	
	(0.020)	(0.026)	(0.003)	
year_blt_decade1950	0.121***	0.077***	0.454***	
	(0.020)	(0.025)	(0.002)	
year_blt_decade1960	0.366***	0.318***	0.622***	
	(0.019)	(0.023)	(0.002)	
year_blt_decade1970	0.285***	0.314***	0.446***	
	(0.022)	(0.026)	(0.002)	
year_blt_decade1980	0.786***	0.717***	0.801***	
	(0.020)	(0.024)	(0.002)	
year_blt_decade1990	0.099***	0.131***	-0.483***	
	(0.032)	(0.036)	(0.003)	
year_blt_decade2000	-0.326***	-0.431***	0.019***	
	(0.023)	(0.027)	(0.002)	
year_blt_decade2010	0.521***	0.401***	0.203***	
	(0.015)	(0.020)	(0.003)	
year_blt_decademissing	0.403***	0.132***	1.12***	
	(0.024)	(0.035)	(0.003)	
year_blt_decadepre-1900	0.019	-0.138***	-0.107***	
	(0.017)	(0.021)	(0.002)	
num_units_bin2-5	-0.348***	-0.306***	1.10***	
	(0.008)	(0.011)	(0.010)	
num_units_bin21-50	-1.38***	-1.35***	0.280***	
	(0.014)	(0.017)	(0.010)	
num_units_bin51+	-2.31***	-2.61***	-0.786***	
	(0.018)	(0.021)	(0.010)	
num_units_bin6-20	-1.28***	-1.16***	0.207***	
	(0.013)	(0.017)	(0.010)	
mean_log_rent_inf_adj		0.447***		
		(0.011)		
portfolio_evict_group_congSmallportfolio			-0.155***	
			(0.002)	
portfolio_evict_group_congLow-evictingportfolio			-0.419***	
			(0.0010)	
portfolio_evict_group_congHigh-evictingportfolio			-0.226***	
			(0.002)	
portfolio_evict_group_cong = Solo				0.026
				(0.281)
portfolio_evict_group_cong = Low-evictingportfolio				0.081
				(0.352)
portfolio_evict_group_cong = High-evictingportfolio				0.213

10.4 Landlord Response to Severe Complaints

The following coefplot shows estimates from a PPML regression of `total_permits` on contemporaneous, lagged (past quarters), and lead (future quarters) severe complaint counts, with PID and year fixed effects. Lead terms serve as pre-trend checks.

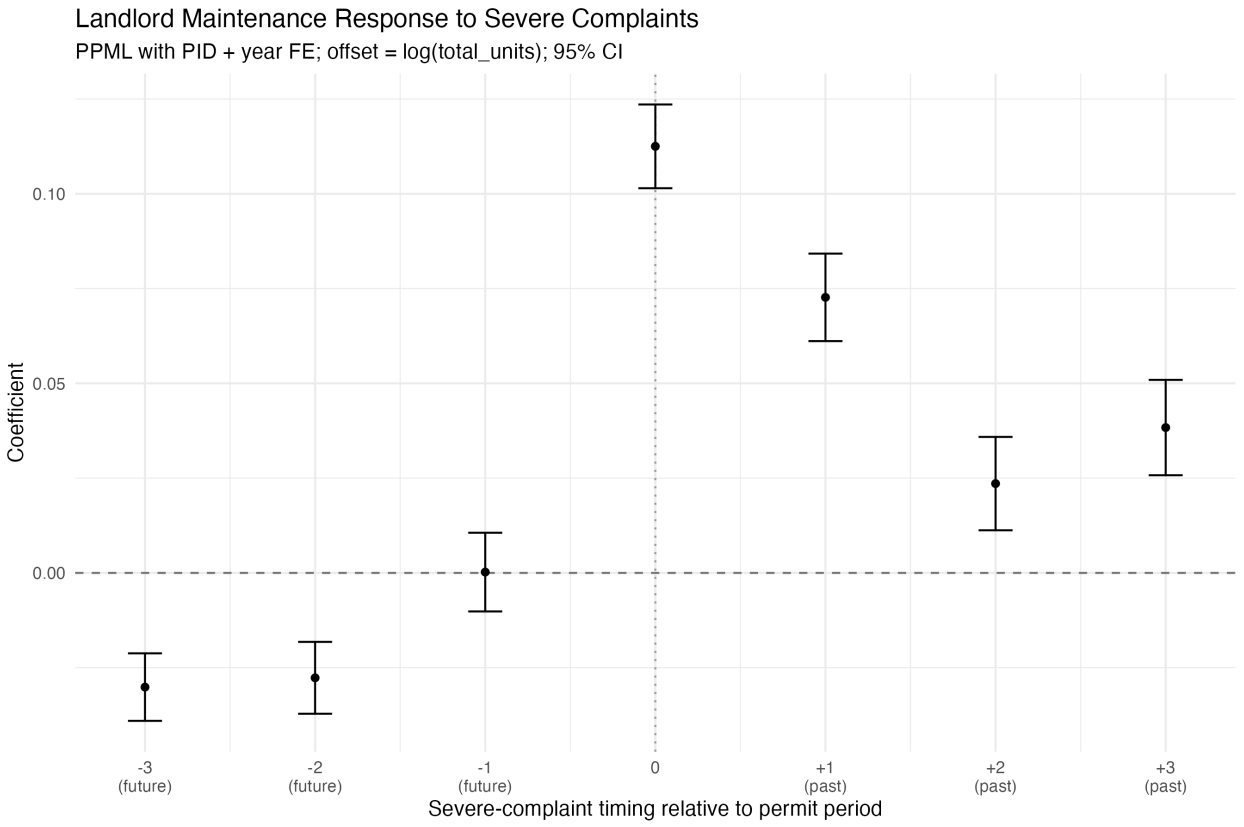

Figure 4: Landlord maintenance response to severe complaints: PPML event-study. Negative timing = complaints occur in future periods relative to permit outcome (pre-trend check); positive timing = past complaints. PID + year FE; offset = log(total_units); 95% CI.

10.5 Landlord Maintenance Response by Portfolio Eviction Group

Same PPML specification as above, estimated separately for each portfolio eviction group. This tests whether **high-evicting-portfolio landlords are less maintenance-responsive** to severe complaints than solo or low-evicting-portfolio landlords.

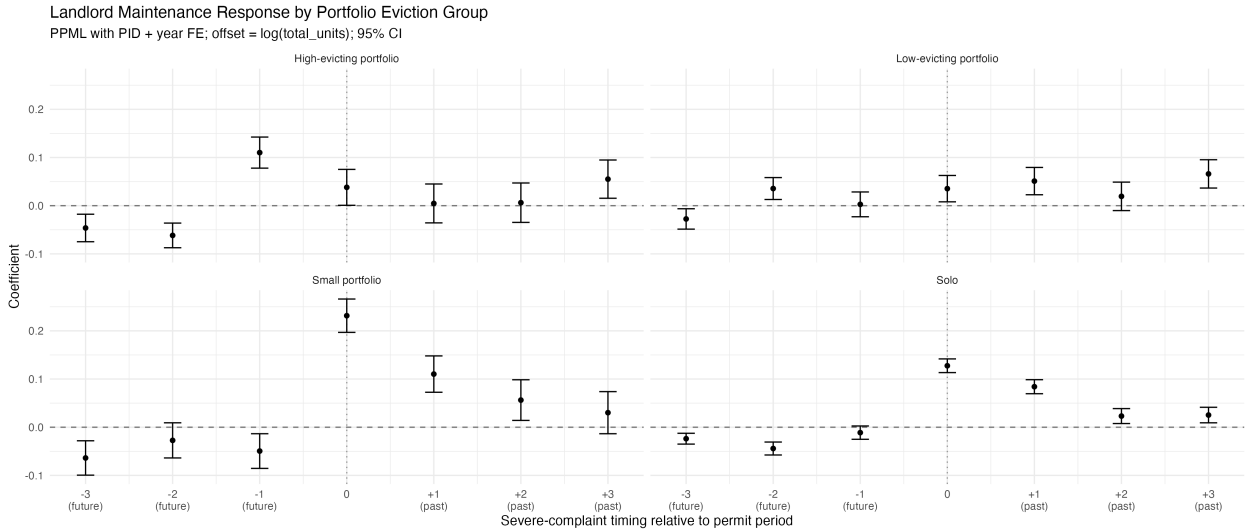

Figure 5: Landlord maintenance response by portfolio eviction group: PPML event-study, faceted by Solo / Low-evicting portfolio / High-evicting portfolio. PID + year FE; offset = log(total_units); 95% CI.

11 Notes

- Raw long-run filing-rate maintenance regressions are moved to the appendix section below.
- This writeup intentionally excludes same-period, targeting, LP-DiD, and complaint-suppression blocks.
- For complaint-suppression specifications and interpretation, use `writeups/comp-statics-annual-race.qmd`

12 Appendix: Raw Filing Robustness

These regressions mirror the baseline maintenance models but replace EB filing intensity with raw long-run pre-2019 filing rate (same 15% high-filing cutoff and same bins), for robustness only.

Dependent Variable:	total_permits			
Model:	(1)	(2)	(3)	(4)
Variables				
above_mean_pct_black_btTRUE	-0.408***	-0.344***	-0.400***	-0.337***
	(0.007)	(0.009)	(0.007)	(0.009)
high_filing_raw	-0.199***	-0.123***		
	(0.013)	(0.014)		
mean_log_med_rent_inf_adj_PID		0.415***		0.399***
		(0.010)		(0.010)
filing_bin_raw = [1,5)")			-0.137***	-0.151***
			(0.010)	(0.011)
filing_bin_raw = [5,10)")			-0.212***	-0.146***
			(0.009)	(0.011)
filing_bin_raw = [10,20)")			-0.162***	-0.088***
			(0.012)	(0.013)
filing_bin_raw = [20,inf)")			-0.296***	-0.204***
			(0.014)	(0.016)
Fixed-effects				
building_type	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
quality_grade	Yes	Yes	Yes	Yes
year_blt_decade	Yes	Yes	Yes	Yes
num_units_bin	Yes	Yes	Yes	Yes
GEOID	Yes	Yes	Yes	Yes
has_rent_data	Yes		Yes	
Fit statistics				
Observations	1,683,222	749,684	1,683,222	749,684
Squared Correlation	0.30726	0.23920	0.30685	0.24032
Pseudo R^2	0.20498	0.25438	0.20546	0.25476
BIC	1,133,791.3	609,927.4	1,133,158.6	609,663.5

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*